

Zelin Li

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Education

- University of Minnesota, Twin Cities**, Bachelor's Degree in Computer Science Sept 2025 – May 2027
• **GPA: 4.0/4.0**
- Beijing University of Posts and Telecommunications (BUPT)**, Artificial Intelligence Sept 2023 – June 2025
• GPA: 3.85/4.0, **Rank: 1/180**

Research Interests

LLM Inference and Post-Training · Zeroth-Order Optimization · Algorithm-System Co-Design · Optimization Theory

Publications & Preprints

Inference-Native Zeroth-Order Optimization.

Zelin Li, Caiwen Ding.
[arXiv:2605.28760](https://arxiv.org/abs/2605.28760) [🔗](#), 2026.

Sample-Stable Subspaces and Mini-Batch Noise in Zeroth-Order Fine-Tuning.

Zelin Li, Caiwen Ding.
Preprint, 2026.

DynamicMem: A Long-Horizon Memory Benchmark in Real-World Settings.

Wenya Xie, Shengming Zhou, **Zelin Li**, et al.
[arXiv:2606.22877](https://arxiv.org/abs/2606.22877) [🔗](#), 2026. Under review at NeurIPS 2026.

Research Experience

Research Assistant, University of Minnesota, Twin Cities Apr 2026 – Present
Advisor: Prof. Caiwen Ding

Inference-Native ZO Systems

- Formulated ZO as candidate-state queries and developed an inference-native execution path that separates algorithmic probe semantics from runtime realization.
- Built the vLLM-based prototype with LoRA-side candidate/update state, lazy and banked updates, packed variable-length scoring, and mixed foreground inference/ZO execution.

ZO Stability and Mini-Batch Noise

- Studied why structured, low-dimensional ZO remains batch-sensitive even after the search geometry becomes highly stable across samples.
- Designed large-batch, fixed-subspace, and matched-direction experiments on OPT-13B and Qwen3 to isolate mini-batch objective noise from search-space variation.

Research Assistant, University of Minnesota, Twin Cities Nov 2025 – Mar 2026
Advisor: Prof. Zirui Liu; worked with Wenya Xie

Long-Horizon LLM Memory Benchmark

- Implemented and evaluated HippoRAG and SimpleMem; contributed to benchmark-data construction and extensive manual auditing of generated data.
- Used audit findings to support data and algorithm revisions, and contributed to experimental analysis and manuscript preparation.

Honors & Awards

National Scholarship, Ministry of Education of China (5/470) 2024
Silver Medal, International Collegiate Programming Contest (ICPC), Asia Shenyang Regional (84/399) 2024
Bronze Medal, CCPC Harbin 2024
Bronze Medal, CCPC Beijing 2024
First Prize, National Olympiad in Informatics in Provinces (NOIP), Shanxi 2021

Technical Skills

Programming: Python, C++ **Frameworks:** PyTorch, vLLM, Hugging Face Transformers **Tools:** Linux, Git